

Reinforcement Learning — From Simulation to Deployment on the Robot Platform Spot

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What is the aim of the thesis?

- Evaluate the development of reinforcement learning policies
- Transfer from simulation to reality
- Leverage Isaac Sim and Isaac Lab for simulation and training
- Establish pipeline from task design to real-world deployment
- Presents two foundational skills:
 - Robust stand-still & Locomotion

Research questions

- **How** can the process of developing an RL policy trained in Isaac Lab for deployment on real Spot hardware be designed, and what is required for **safe** development and deployment of such a policy?
- How does the trained policy transfer from simulation to reality, reducing the sim-to-real gap?
- What is required for iteratively improving the policy?

The robot dog Spot

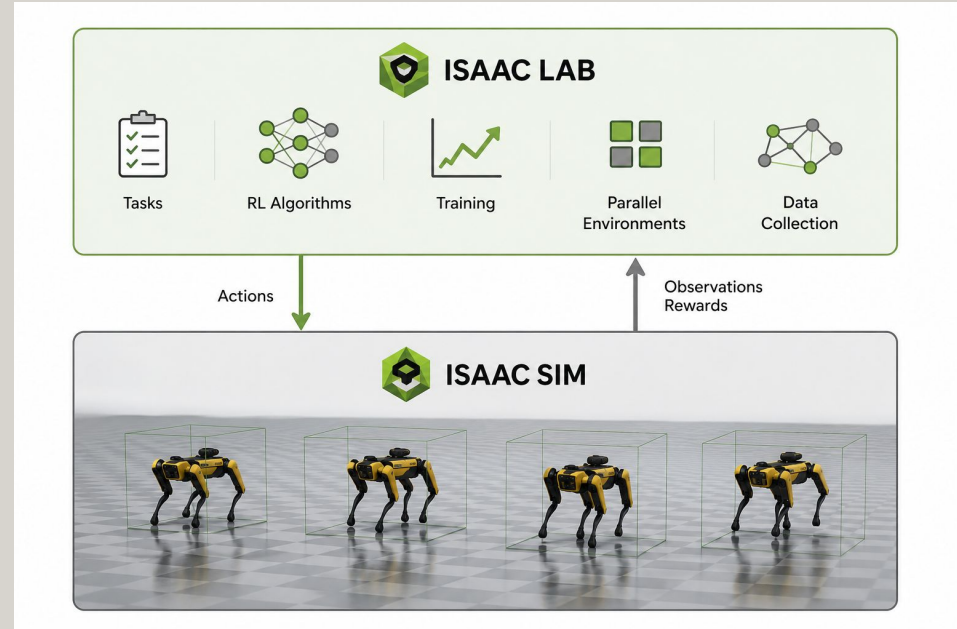
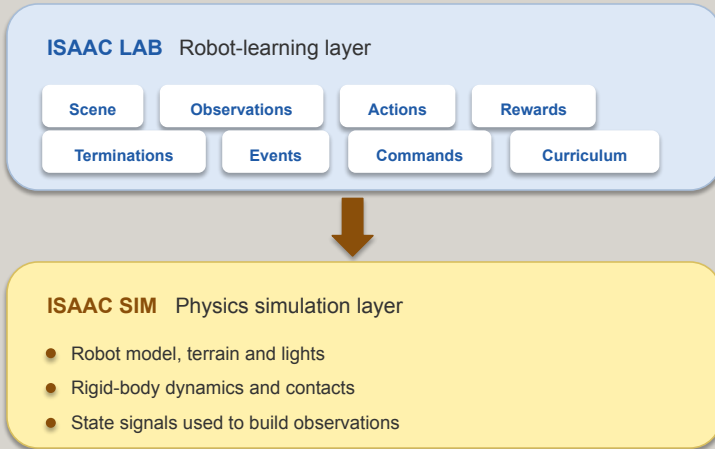
- Quadruped with 12 leg joints
- Manipulator with 6 arm joints + 1 gripper
- Attached with a NVIDIA Jetson AGX Orin
- Uses Spot RL Researcher Kit for RL integration
 - Enables communication between Spot and RL policies
 - Access to joint-level control
- Heavy and powerful platform requires careful operation
- Potential risks for operator and hardware during use
- Training in simulation is advantageous



Isaac Sim and Isaac Lab

The structure

Isaac Lab defines what should be learned;
Isaac Sim executes one physics step at a time.



Isaac Lab RL environment structure

Observations

input vector given to the policy

Rewards

numerical feedback shaping behaviour

Actions

joint-position targets for leg and arm

Scene

robot, ground, lights, contact sensors

Events

domain randomization and disturbances

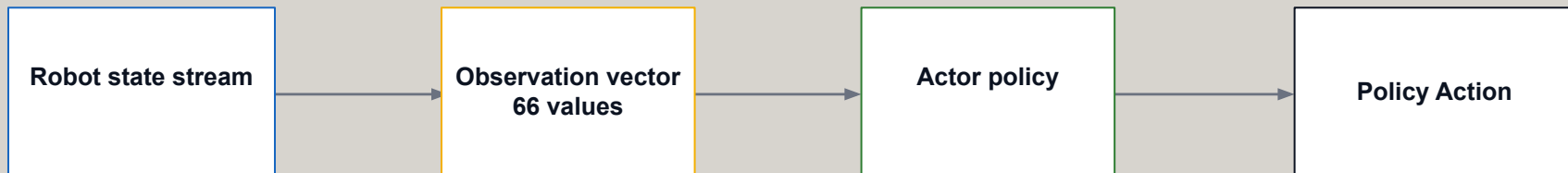
Terminations

falls, unsafe contacts or timeouts

Observation component	Dimension	Purpose
Base linear velocity	3	Describes translational body motion.
Base angular velocity	3	Describes rotational body motion.
Projected gravity	3	Gives body orientation relative to gravity.
Joint positions	19	Gives current joint configuration.
Joint velocities	19	Gives joint motion state.
Previous action	19	Helps stabilize policy behaviour over time.
Total	66	Full policy observation vector.

Default observation vector: 66 values; task commands expand it to 69 values.

How the policy controls the robot



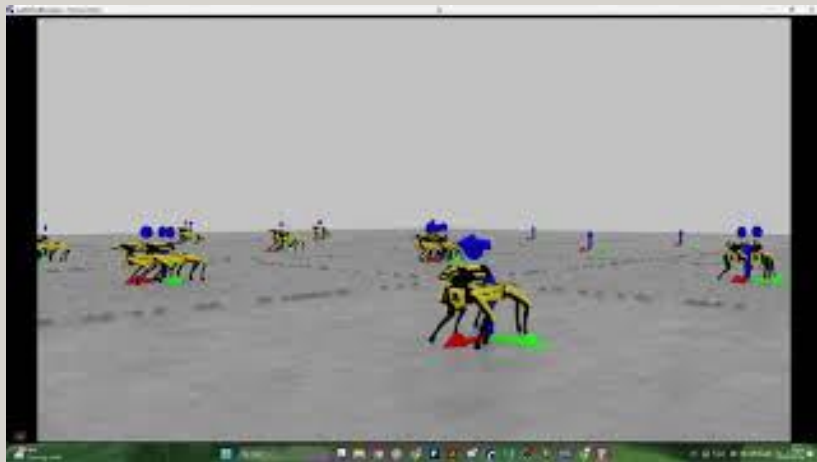
$$q_d = q_{default} + a_{scale} \cdot a_{\pi}$$



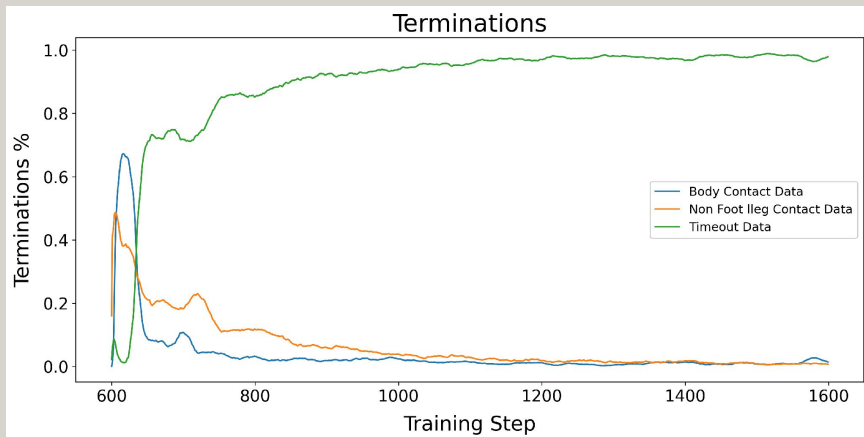
Task 1: Robust standing

Maintain balance, stay still, and recover to a predefined standing pose after disturbances.

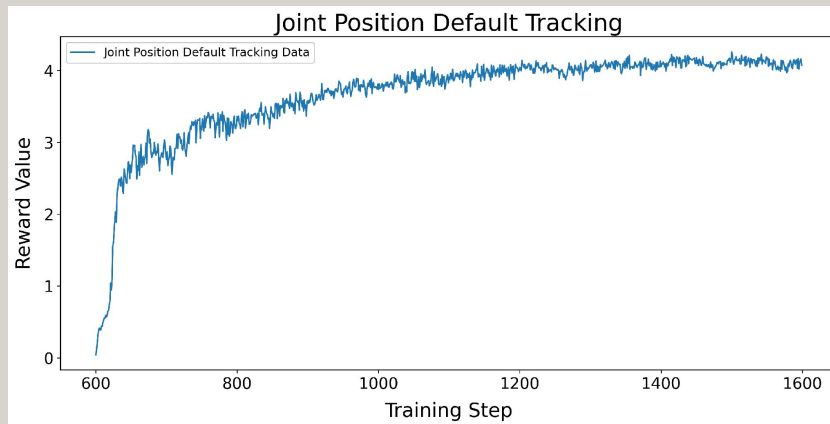
- Domain randomization
- Reward structure



Robust standing learning recovery behaviour



Recovery emerge:
~650



Survival rate:
97%

Task 2: Locomotion

- Move in a commanded direction
- Randomized direction and magnitude of the command
- Additional domain randomized perturbations

Three approaches were tested

Two-phase curriculum

Phase 1: Gaiting phase
Phase 2: Tracking phase

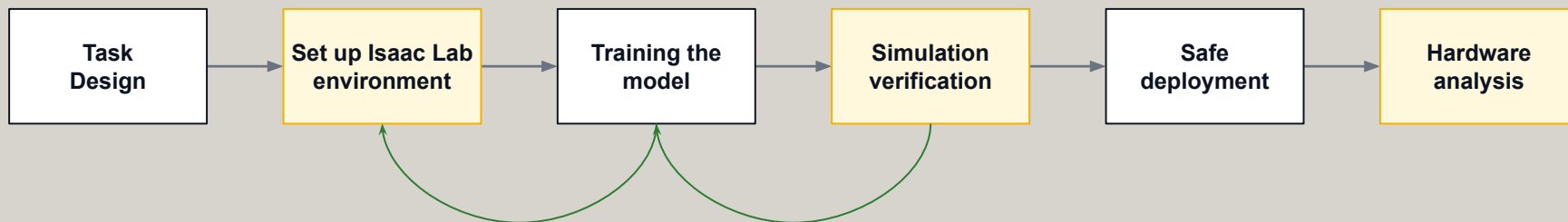
Policy blending

Combine with standing.
Requires fine tuning

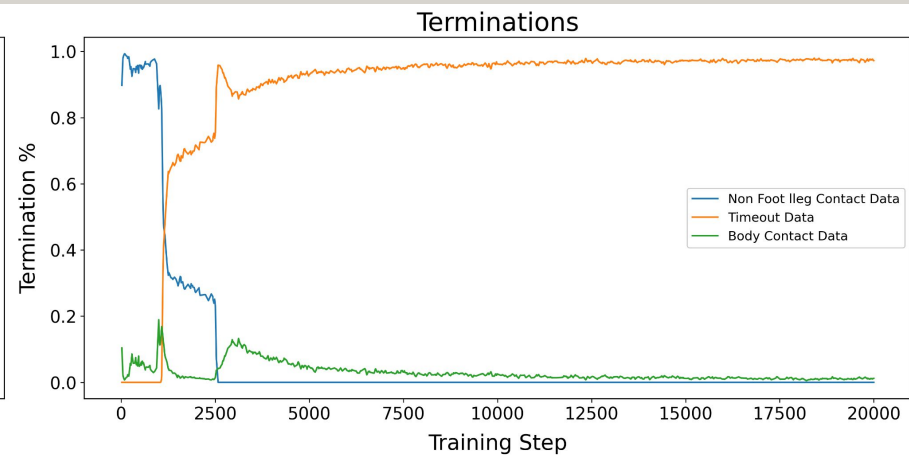
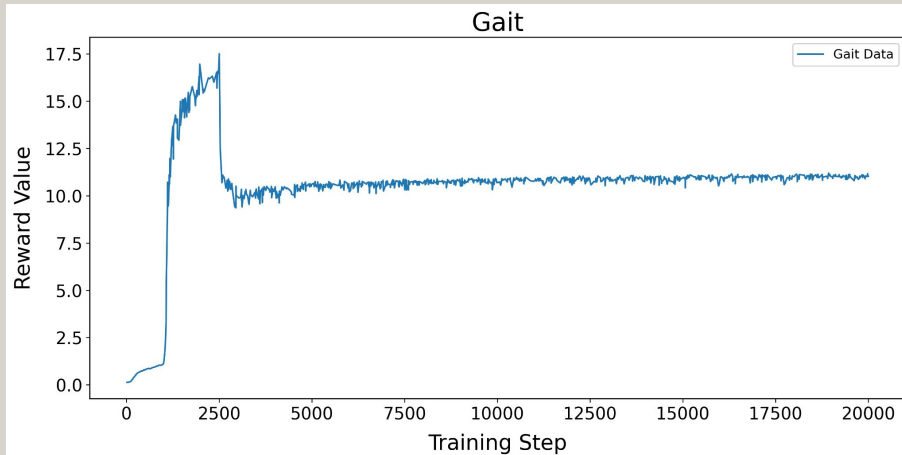
Warm-starting

Adapt locomotion policy for
spot with arm attached

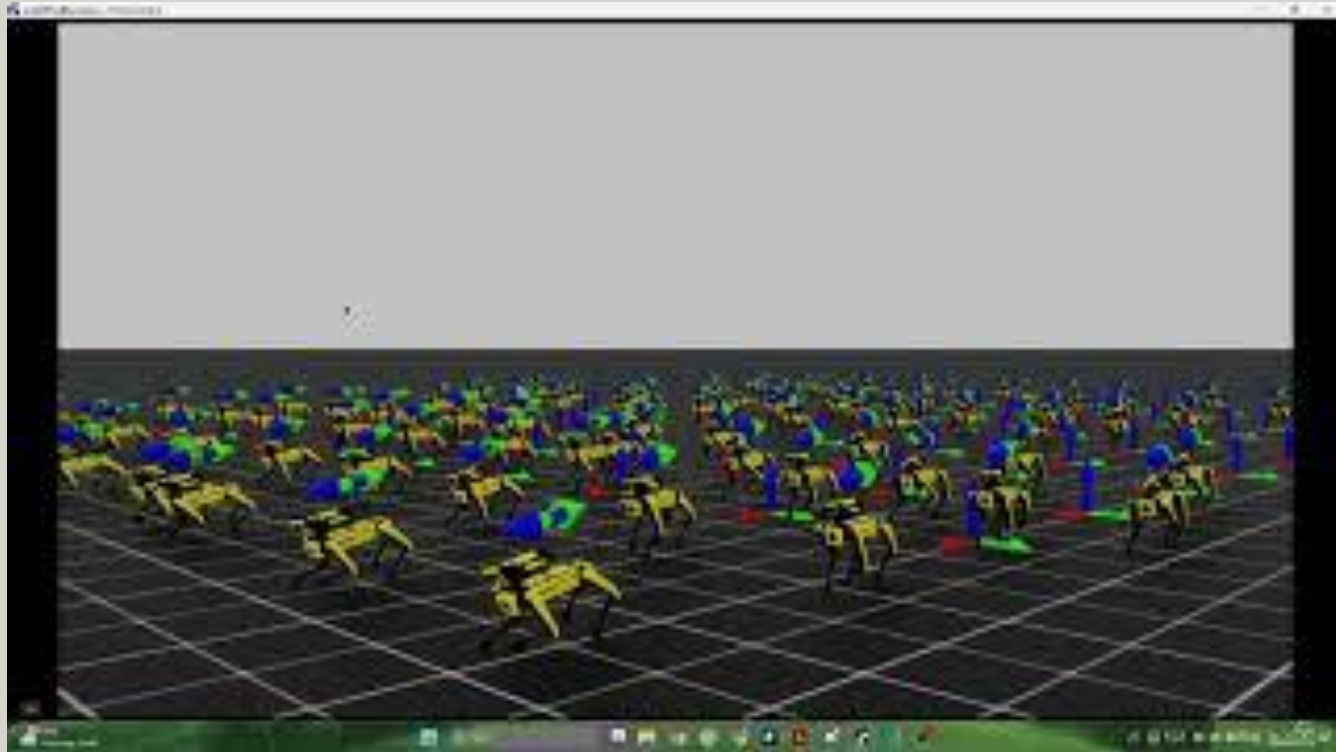
The Development Loop



Two-phase locomotion training

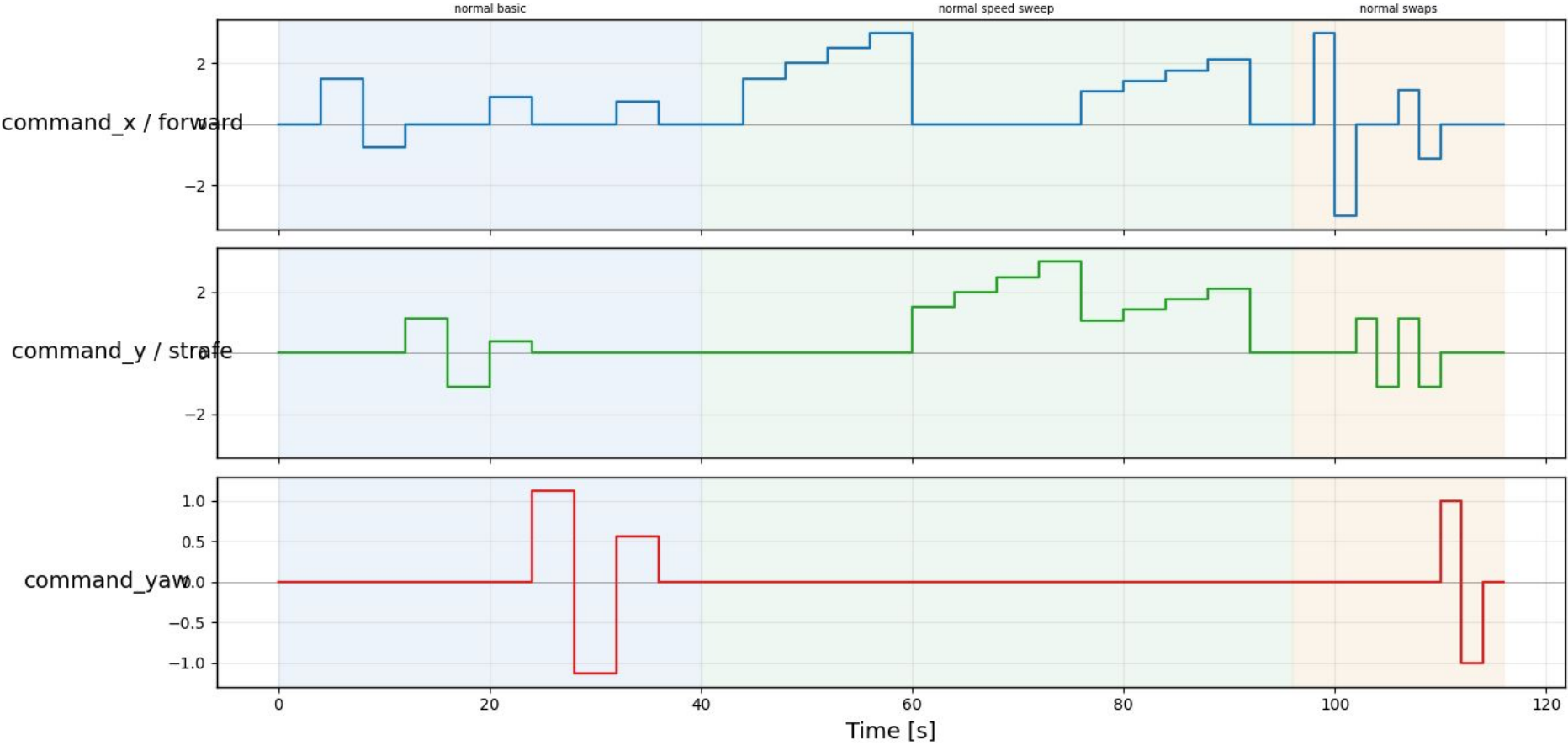


Simulated locomotion test suites



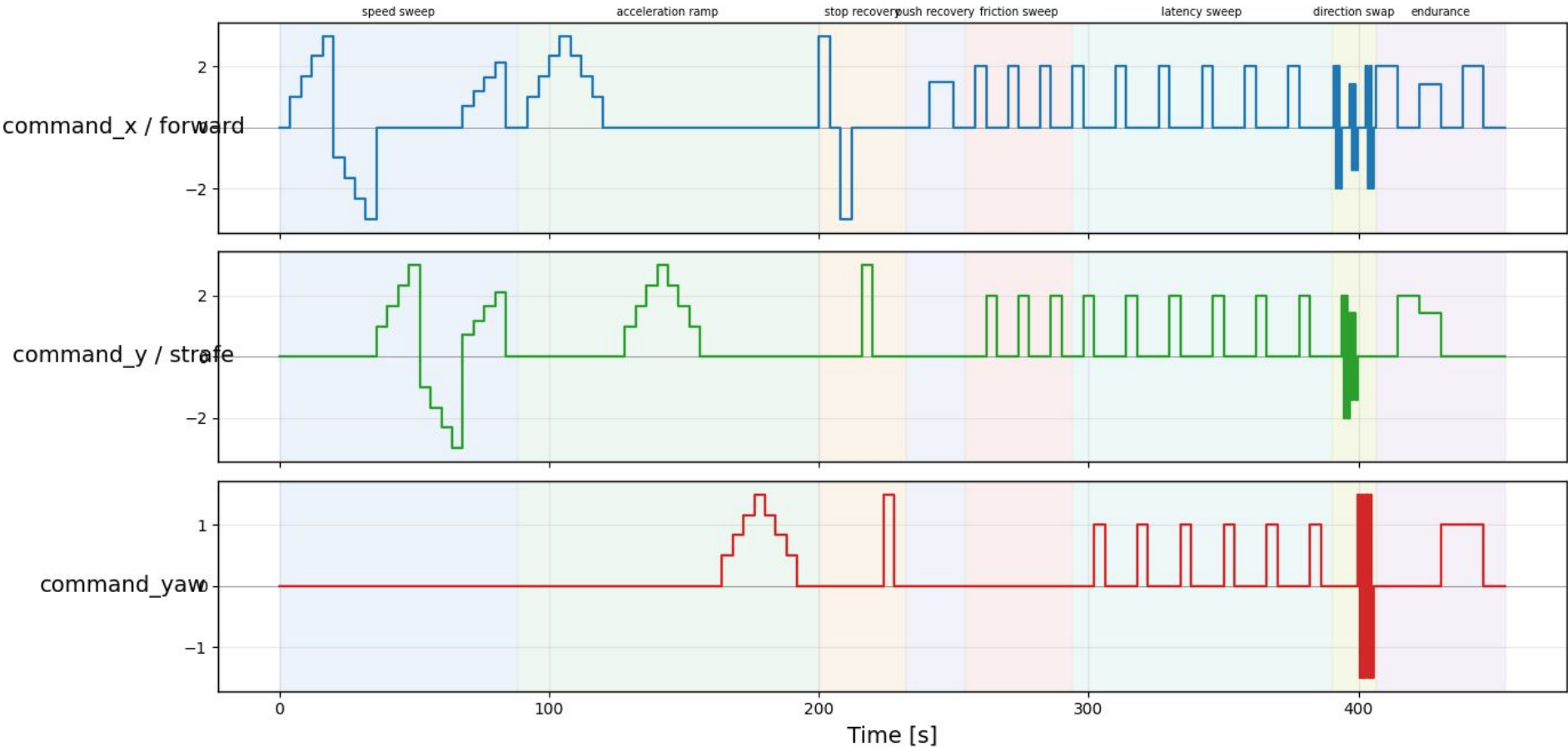
normal suite command profile

Total cycle: 116.0s (1.9 min)



stress suite command profile

Total cycle: 454.5s (7.6 min)



Policy	Task success [%]	Lin. RMSE [m/s]	Yaw RMSE [rad/s]	Tilt mean [rad]	Foot slip [m/s]	Torque mean [Nm]
NVIDIA baseline	97.3%	0.434	0.119	0.071	0.264	11.43
NVIDIA fine-tune	100.0%	0.427	0.126	0.057	0.259	11.10
Two-phase	100.0%	0.399	0.135	0.036	0.297	20.15
Two-phase blended	99.9%	0.370	0.136	0.034	0.266	18.51

Table 4.1 Normal-suite locomotion policy comparison for 15872 simulated tasks, without self collision.

Policy	Task success [%]	Lin. RMSE [m/s]	Yaw RMSE [rad/s]	Tilt mean [rad]	Foot slip [m/s]	Torque mean [Nm]
NVIDIA baseline	93.4%	0.464	0.245	0.079	0.285	11.60
NVIDIA fine-tune	95.0%	0.459	0.282	0.077	0.275	11.16
Two-phase	97.2%	0.462	0.273	0.049	0.340	20.53
Two-phase blended	96.9%	0.424	0.282	0.047	0.295	18.23

Table 4.2 Stress-suite locomotion policy comparison of 64512 simulated tasks.

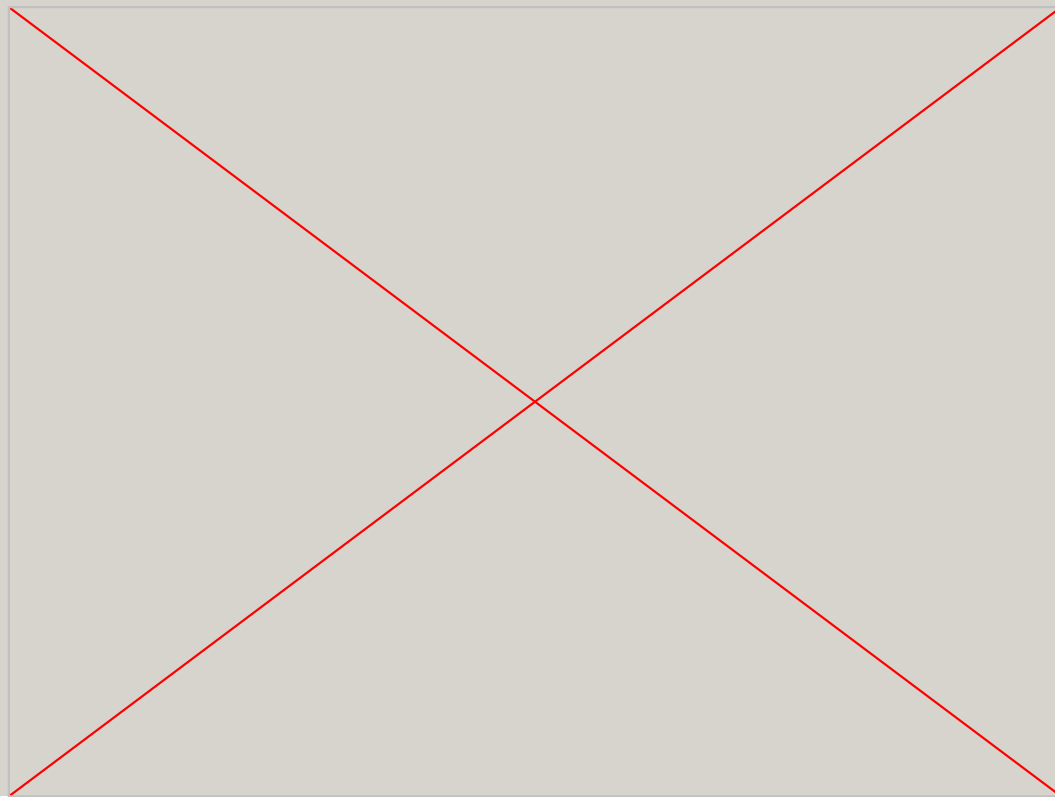
Safe policy deployment

- Sim-to-real gap
- Safety is crucial
- Harness suspension
 - Deploy policies on Spot while airborne
 - Catch spot in the case of balance loss
 - Reasonability of the resulting behaviour
- Specified joint control
- Scaled policy output
- Evaluate policy behaviour in simulation





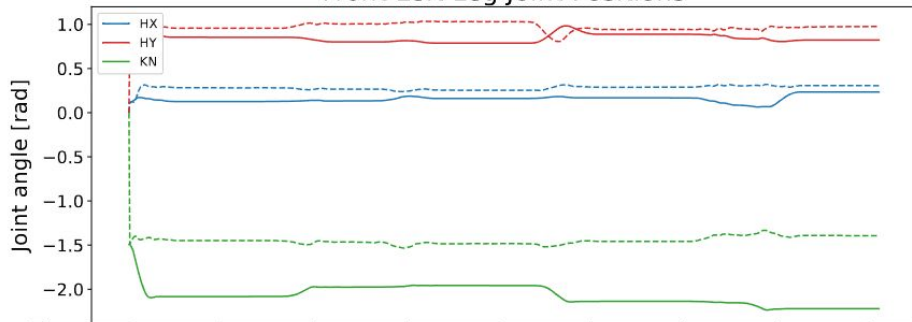
Standing deployment



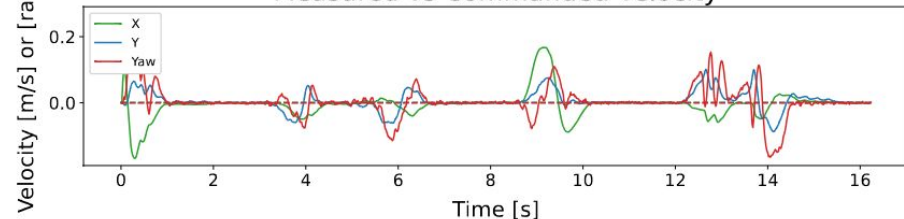
Standing deployment

HARDWARE

Front Left Leg Joint Positions

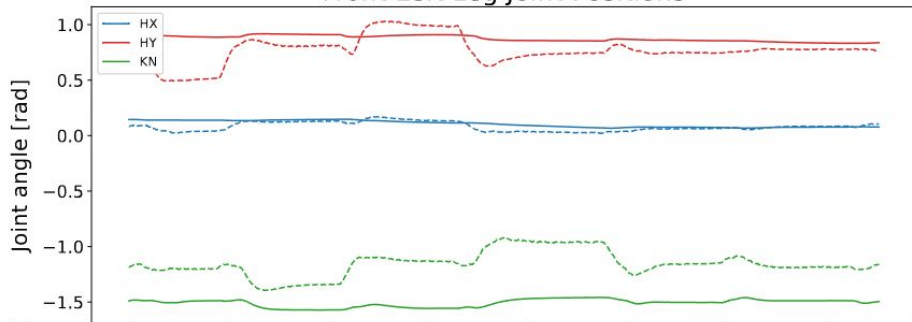


Measured vs Commanded Velocity

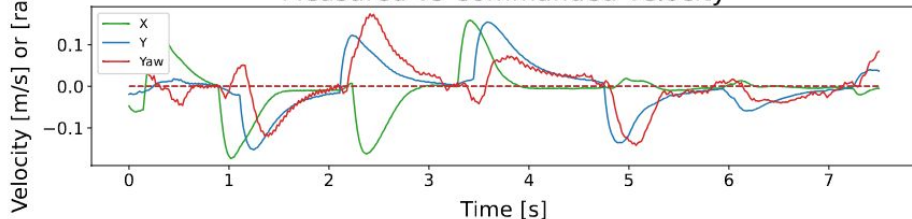


SIMULATION

Front Left Leg Joint Positions



Measured vs Commanded Velocity



Solid line — measured joint position/measured velocity. Dashed line — policy output/commanded velocity

Policy recover from small pushes

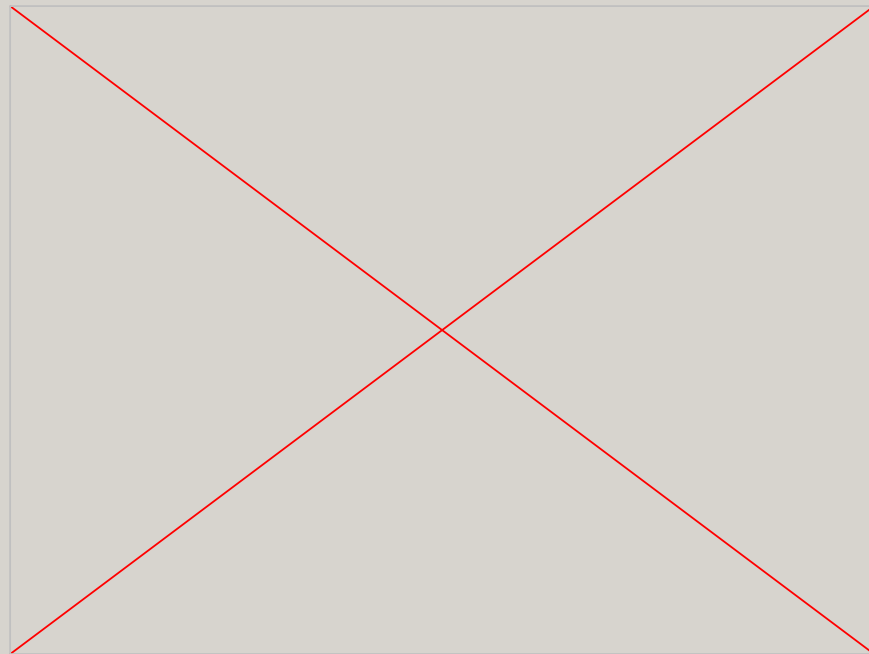
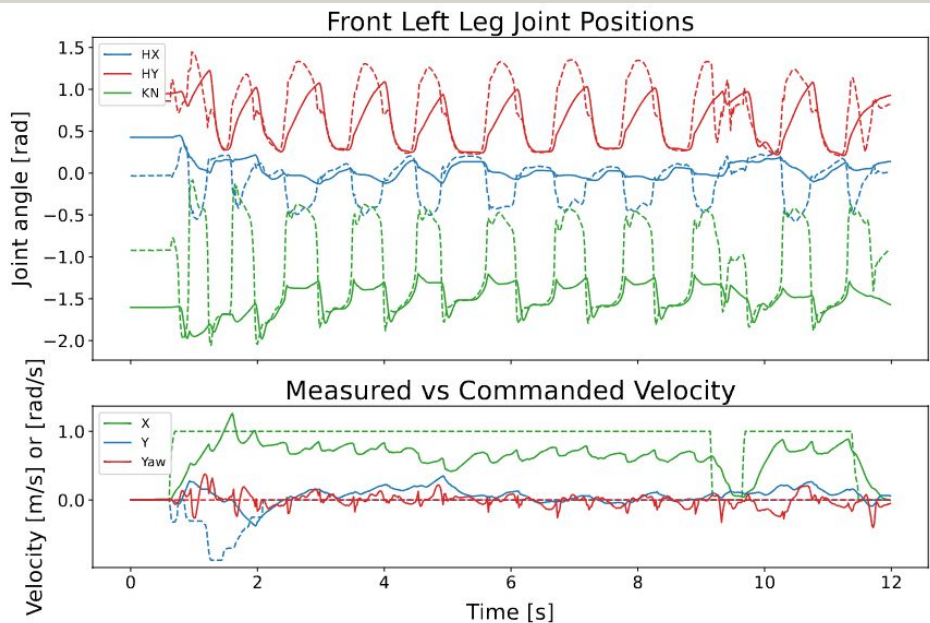
Static error between policy output and position of knee joint — influence from the underlying PD-controller

Error is larger during deployment — potential sim-to-real gap?

Locomotion deployment

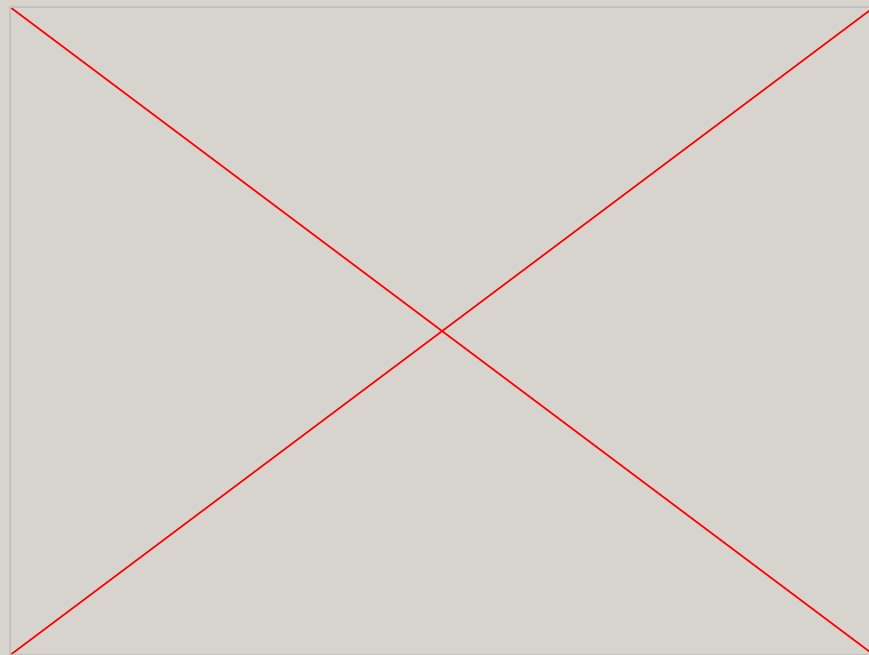
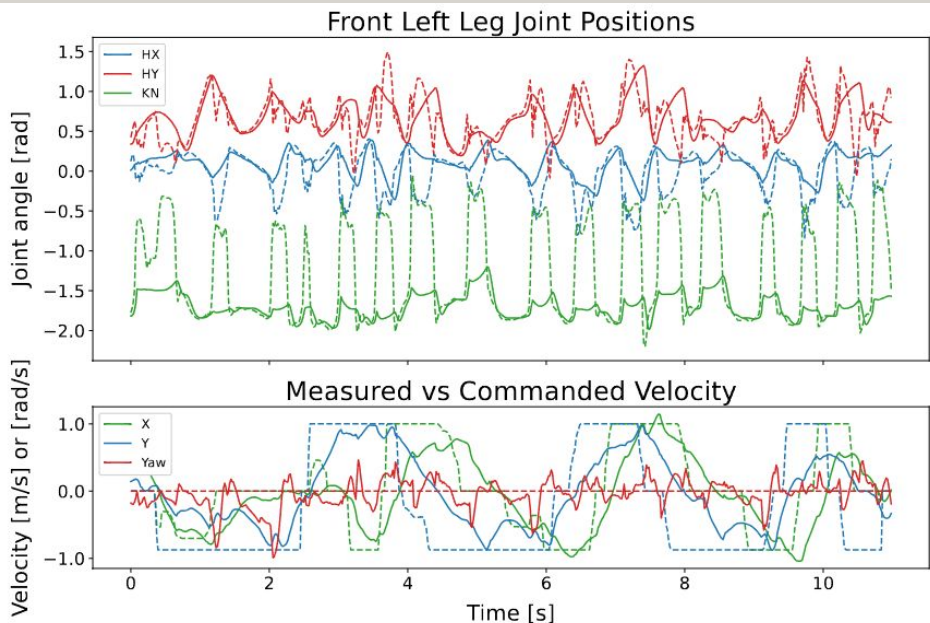
- 3 different policies deployed on hardware
- Version before introduction of two-phase
 - Bad command tracking - not able to take a step
- The two-phase policy
 - Very aggressive - deemed unsafe
- Fine-tuned policy
 - Smooth movements and viable for locomotion
 - Used for further testing of robustness and capabilities
- Further testing include
 - Continuous walking
 - Different directions
 - Robustness against external pushes

Locomotion deployment: Continuous walking



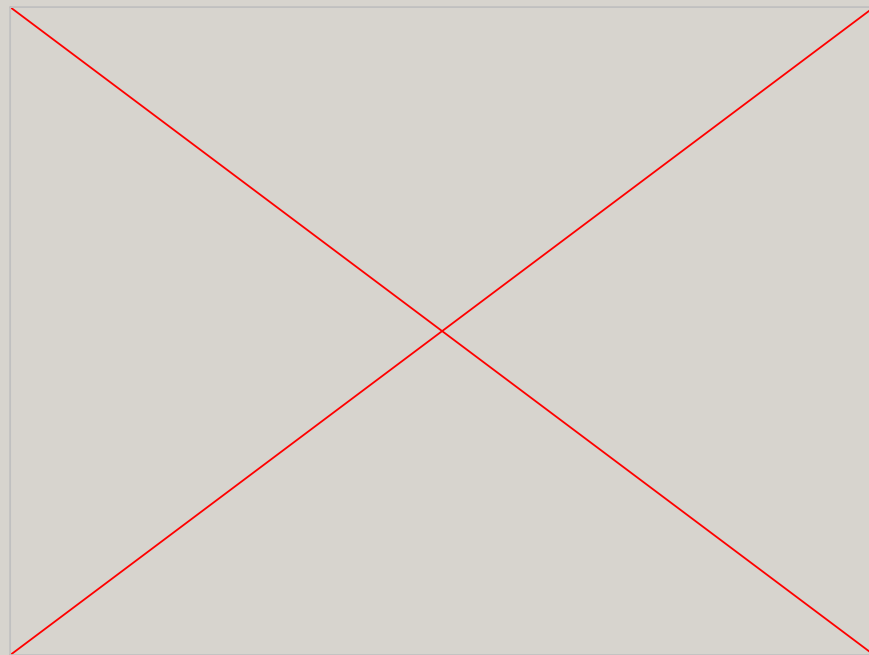
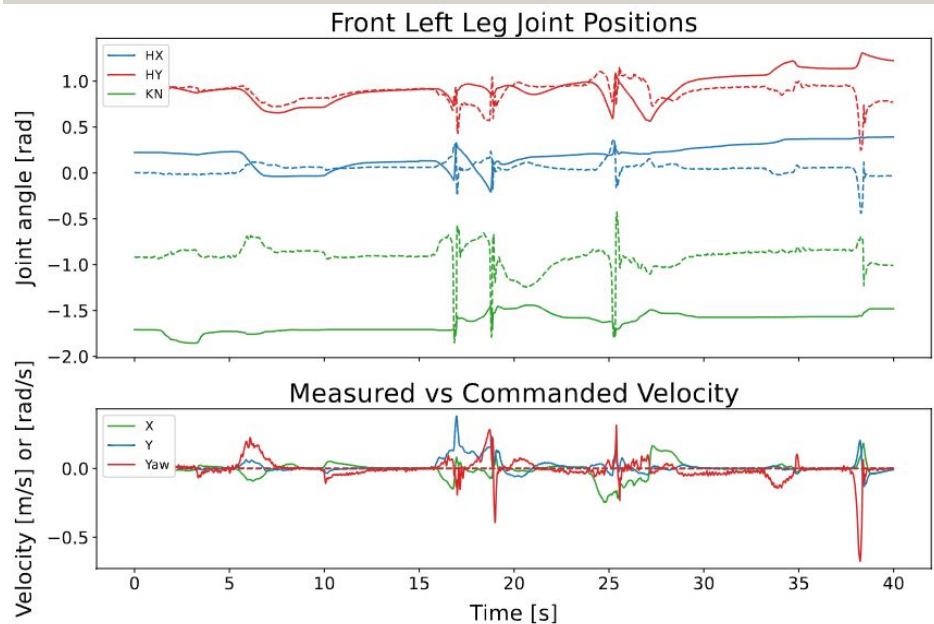
Solid line — measured joint position/measured velocity. Dashed line — policy output/commanded velocity

Locomotion deployment: Diverse walking



Solid line — measured joint position/measured velocity. Dashed line — policy output/commanded velocity

Locomotion deployment: External pushes



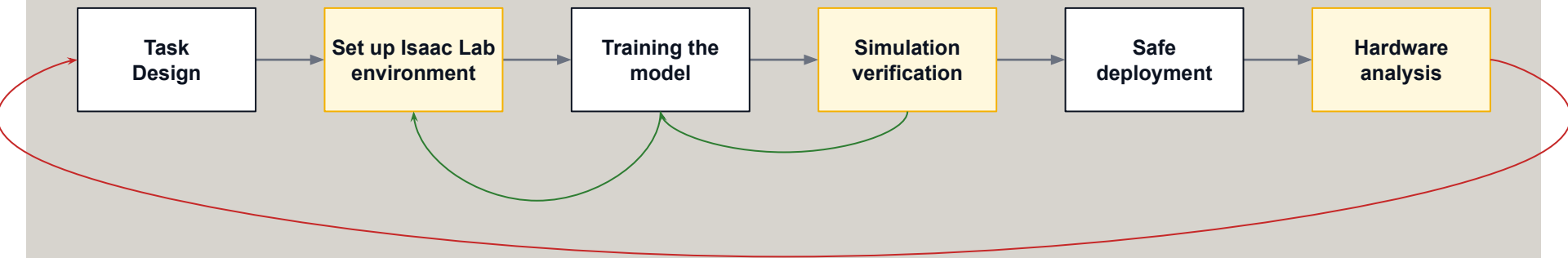
Solid line — measured joint position/measured velocity. Dashed line — policy output/commanded velocity

Concluding thoughts

- Presented a development method for designing and, training reinforcement learning policies and deployment on Spot
- The iterative and exploratory nature
- Presented safety practises for this pipeline
- Observed transfer from simulation to reality

Future work

- Utilize the data gathered from deployment to improve modeling
- experiment without the safety harness
- Extend from locomotion to manipulation
- Integrate policy control into a larger context



Questions?